

GHRCEM WAGHOLI PUNE
SY CSE AI,AI ML,CS & DS
QUESTION BANK FOR CAE 3
SUBJECT: DMGT (UBSL205)

Sr. no	Question	Unit	BL	CO	PO
1	A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, one in Java and Python, five in C# and Python, and just one programmer is proficient in all three languages above. Determine the number of computer programmers that are not proficient in any of these three languages	5	3	2	1
2	There are 350 farmers in a large region. 260 farm beetroot, 100 farm yams, 70 farm radish, 40 farm beetroot and radish, 40 farm yams and radish, and 30 farm beetroot and yams. Let B, Y, and R denote the set of farms that farm beetroot, yams and radish respectively. Determine the number of farmers that farm beetroot, yams, and radish	5	3	2	1
3	In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the number of families which buy i)Number of families which buy all three newspapers. ii)Number of families which buy newspaper A only iii)Number of families which buy newspaper B only iv)Number of families which buy newspaper C only v)Number of families which buy None of A, B, C	5	3	2	1

4	Solve $a_n = a_{n-1} + 2a_{n-2}$, $a_0 = 2, a_1 = 7$	5	5	3	1
5	Solve $a_n = -a_{n-1} + 4a_{n-2} + 4a_{n-3}$, $a_0 = 8, a_1 = 6, a_2 = 26$	5	5	3	1
6	Solve $F_n = 5F_{n-1} - 6F_{n-2}$, $F_0 = 1, F_1 = 4$	5	5	3	1
7	Solve $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$, $a_0 = 1, a_1 = -2, a_2 = -1$.	5	5	3	1
8	Solve $a_n = 8a_{n-2} - 16a_{n-4}$, for $n \geq 4$ $a_0 = 1, a_1 = 4, a_2 = 28, a_3 = 32$	5	5	3	1
9	How many friends must you have to guarantee that at least five of them have their birthday in the same month?	5	2	2	2
10	If eight persons are chosen from any group, show that at least two of them will have the same birthday	5	2	2	2

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SUBJECT INCHARGE

HOD